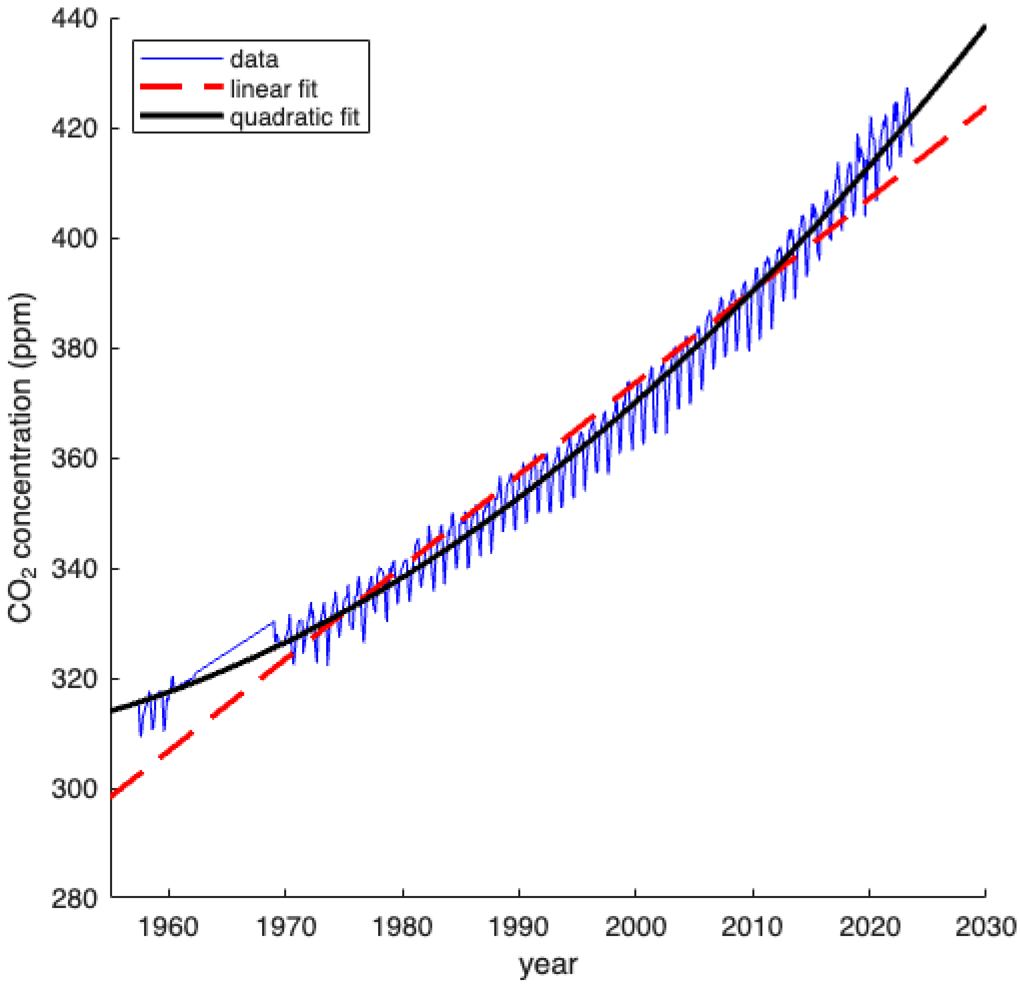
```
% ch10_co2.m
% creates fig 10.1 (with linear and quadratic fits)
% and residual plots of fig 10.3
close all; clearvars; clc;

% load data
% https://gml.noaa.gov/webdata/ccgg/trends/co2/co2_mm_mlo.txt
% edit with text editor: Notepad (windows) / TextEdit (mac) / ...
% - note what each column means
% % To read CSV file and tell MATLAB to skip over lines 1-57 (data starts at
line 58 of text file)
filename = 'monthly_ljo.csv';
c = readtable(filename, 'NumHeaderLines', 57);

t = c.Var1;  % year
mo = c.Var2; % month
x = c.Var4;  % decimal date
y = c.Var5;  % CO2

% Below, we use the rows of the CO2 column to find and remove (entire) rows
% that contain NaN values (the entire row will not be used).
keep = ~isnan(y);
t = t(keep);
mo = mo(keep);
x = x(keep);
y = y(keep);
n = length(x);

% simple linear regression
xbar = mean(x);
ybar = mean(y);
xybar = mean(x.*y);
x2bar = mean(x.*x);
m = (xybar -xbar*ybar)/(x2bar - xbar^2)
b = ybar - m*xbar
yfit = m*x+b; % predicted y

% for plotting the fitting function
nplt = 200; % points for ploting line
xplt = linspace(1955,2030,nplt); % range of x
yplt = m*xplt + b; % fitting function

% coefficient of determination
r = y - yfit; % residuals
Rsq = 1 - norm(r)^2/norm(y-ybar)^2 % coef of determination

% quadratic fit:
pcoef = polyfit(x,y,2); % pcoef(1) = alpha(3), etc

% print monthly residuals
fprintf('quadratic fitting function \n') % formatted print, \n = new line
% format "%f" for floating point number
```

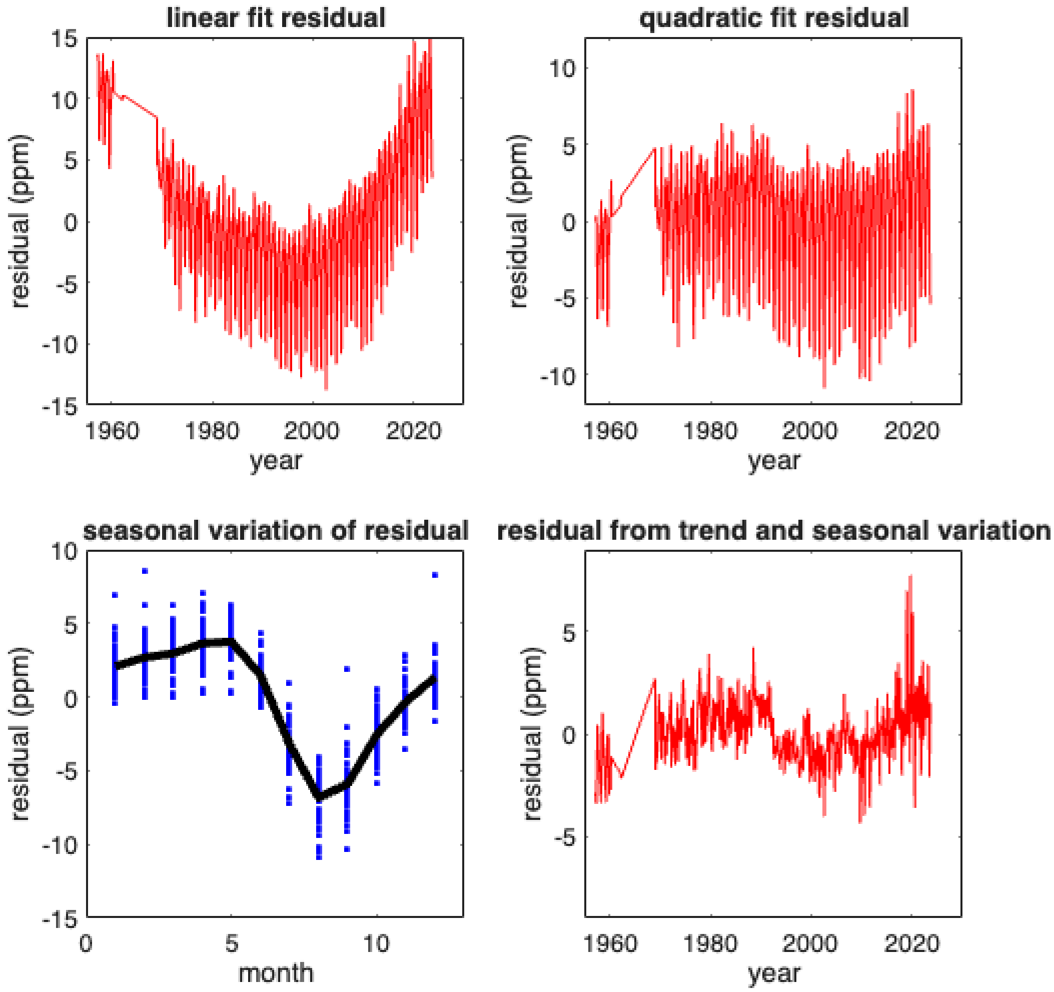
```
fprintf('%f + %f x + %f x^2 \n',pcoef(3),pcoef(2),pcoef(1))
% could use xshift = x - 1950 to get better scaled coefficients

y2fit = pcoef(3)+ pcoef(2)*x + pcoef(1)*x.^2; % predicted y
r2 = y - y2fit; % residuals
Rsq2 = 1 - norm(r2)^2/norm(y-ybar)^2 % coef of determination
% fitting function
y2plt = pcoef(3)*ones(1,nplt) + pcoef(2)*xplt + pcoef(1)*xplt.^2;

% fig 10.1 but with linear and quadratic fits
figure(1)
hold on
plot(x,y,'b-')
plot(xplt,yplt,'r--','LineWidth',2)
plot(xplt,y2plt,'k-','LineWidth',2)
xlim([1955 2030])
legend('data','linear fit','quadratic fit','Location','NorthWest')
xlabel('year'); ylabel('CO_2 concentration (ppm)')
hold off

% seasonal variation
% extract residual for each month and calculate average
nmo = zeros(1,12); % bin for counting number of times given month has data
r2mo = zeros(1,12); % bin for cumulative residual for given month
for i=1:n
    imonth = mo(i); % extract month
    nmo(imonth) = nmo(imonth)+1; % add to month count bin
    r2mo(imonth) = r2mo(imonth)+r2(i); % add to cumulative residual
end
r2mo = r2mo./nmo; % average residual for each month

% print monthly residuals
fprintf('monthly variation of residuals \n') % formatted print, \n = new line
for k=1:12
    fprintf('%i \t %f \n',k,r2mo(k)) % use "%i" for integer, "%f" for float
end

% find residuals relative to the monthly average
for i=1:n
    imonth = mo(i); % extract month
    r3(i) = r2(i) - r2mo(imonth); % new residual
end
Rsq3 = 1 - norm(r3)^2/norm(y-ybar)^2 % coef of determination

% plots of residuals
figure(2)
subplot(2,2,1)
    plot(x,r,'r-')
    title('linear fit residual')
    xlim([1955 2030])
    xlabel('year'); ylabel('residual (ppm)')
subplot(2,2,2)
    plot(x,r2,'r-')
    title('quadratic fit residual')
```

```
    xlim([1955 2030]); ylim([-12 12]);
    xlabel('year'); ylabel('residual (ppm)')
subplot(2,2,3)
     plot (mo,r2,'b.')
     hold on
    plot(1:12,r2mo,'k-','LineWidth',3);
    title('seasonal variation of residual')
    xlabel('month'); ylabel('residual (ppm)')
    xlim([0 13])
    hold off
subplot(2,2,4)
    plot(x,r3,'r-')
    title('residual from trend and seasonal variation')
    xlim([1955 2030]); ylim([-9 9]);
    xlabel('year'); ylabel('residual (ppm)')


m =

    1.6731


b =

  -2.9726e+03


Rsq =

    0.9590

quadratic fitting function
51865.985508 + -53.365437 x + 0.013809 x^2

Rsq2 =

    0.9825

monthly variation of residuals
1      2.077340
2      2.680886
3      2.955059
4      3.641358
5      3.753366
6      1.476283
7      -3.219315
8      -6.837133
9      -5.894946
10      -2.545897
11      -0.322033
12      1.352863

Rsq3 =
```

```
0.9974
```



data
linear fit
quadratic fit
440
420
400
380
360
340
320
300
280
CO2 concentration (ppm)
1960
1970
1980
1990
2000
2010
2020
2030
year



Published with MATLAB® R2025b